Department Overview

- **Department Name:** Department of Mining Engineering
- **Faculty:** Faculty of Engineering & Architecture (established in 2004)
- **Location:** Takatu Campus (named after the Takatu Mountain), Quetta, Balochistan.
- **History:** The Department of Mining Engineering was established in 2008. Since its inception, a total of 8 batches have graduated and are serving across various public and private sector mining organizations.
- **Accreditation:** Fully accredited by the **Pakistan Engineering Council (PEC)**.
- **Core Focus:** Cultivating technical competence and applied research to resolve real-time issues of the mining industry. The curriculum focuses heavily on environmentally friendly mineral extraction, engineering professional ethics, workplace safety, and sustainable development.
- **Future Ambitions:** Enhancing corporate and academic outreach by establishing sustainable linkages with local and international mining organizations.

Mission

The department of Mining Engineering is striving for a high-quality undergraduate program supported by a compatible contemporary curriculum, applied research, and provision of technical competence to the graduates possessing skills to contribute to the Mining industry of Pakistan in general and Balochistan in particular with an environment-friendly approach.

Program Educational Objectives (PEOs)

Civil and mining engineering graduates are expected to demonstrate the following attributes 3 to 5 years post-graduation:

1. Recognize, investigate, and address engineering problems related to the exploitation of mineral deposits of Pakistan/Balochistan by employing modern techniques.
2. Exhibit a high level of technical competence, applying research and problem-solving skills to produce solutions in the field of Mining Engineering.
3. Exercise professional excellence utilizing interpersonal skills focused on workplace safety, environmental sustainability, and professional engineering ethics.
4. Pursue lifelong learning by accepting practical challenges in the mining engineering field.

1. Bachelor of Science in Mining Engineering (BS Mining)

Admission Requirements

- **Academic Background:** F.Sc (Pre-Engineering) / ICS with Mathematics & Physics from a recognized board (or equivalent qualification) with at least **60% marks**, OR a Diploma of Associate Engineer (DAE) in a relevant field with a minimum of **60% marks**.

Degree Requirements

- **Total Credit Hours:** 136
- **Total Courses:** 45 – 49
- **Minimum Academic Standing:** CGPA $\ge$ 2.0
- **PLO Requirement:** Mastery of the 12 expected standard engineering graduate attributes (Engineering Knowledge, Problem Analysis, Modern Tool Usage, etc.).

Curriculum Plan (Program Schema by Semester)

Note: Credit hour structures denote (Theory + Lab).

Semester	Course Code	Course Title / Subject	Theory (CH)	Lab (CH)	Total CH
1st	MATHP-120	Analytic Geometry & Calculus	3	0	**17**
	CHE-102 & 102L	Applied Chemistry	3	1	
	PHY-205 & 205L	Applied Physics	3	1	
	ENGG-201(MINE) & L	Engineering Drawing & Graphics	1	2	
	MINE-210	Mining Engineering Fundamentals	3	0	
2nd	MATHP-111	Linear Algebra	3	0	**18**
	ENGG-215	Applied	3	1	

	& 215L	Thermodynamics			
	GEOE-201 & 201L	Applied Geology	3	1	
	HUM-101	Islamic Studies / Ethics	2	0	
	HUM-364 / HUM-206	English Composition & Comp. / Professional Ethics	3 / 2	0 / 0	
3rd	MINE—	Economic Minerals of Pakistan	3	0	**17**
	ENGG-205	Engineering Mechanics	3	0	
	ENGG-216 & 216L	Basic Electrical Technology	3	1	
	ENGG-206 & 206L	Fluid Mechanics	3	1	
	MATHA-214	Differential Equations	3	0	
4th	HUM-268 /	Communication Skills /	3 / 2	0 / 1	**18**

	CS-112L	Computer Programming & Lab			
	MINE-209 & 209L	Mine Surveying	3	1	
	HUM-102 ENGG	Pakistan Studies	2	0	
	GEOE-206 & 206L	Mechanics of Material	3	1	
	STAT-102 MINE	Probability & Statistics	2	0	
5th	HUM-277 / MATHP-272	Report Writing Skills / Numerical Method in Computing	2 / 2	0 / 0	**18**
	GEOE-207 & 207L	Structural Geology	3	1	
	MINE-302 / MINE-307 & L	Mineral Exploration / Explosives Engineering	3 / 3	0 / 1	
	MGMT-201	Entrepreneurship	3	0	

6th	MINE-315	Surface Mine Design	3	0	**17**
	MINE-401 & 401L	Rock Mechanics	3	1	
	MINE—	Mineral Processing - I	3	1	
	MINE-303	Mine Power, Drainage & Material Handling	3	0	
	MINE—	Computer Application s in Mining Engineering	2	1	
7th	MINE-313 / MINE-316 & L	Mining Law / Mine Ventilation	2 / 3	0 / 1	**18**
	MINE-310 & 310L	Mineral Processing – II	3	1	
	MINE-454L	Senior Design Project - I (FYDP-I)	0	3	

	ECON-105 / MGMT-304	Engineering Economics / Organizational Behavior	2 / 3	0 / 0	
8th	MINE-455L	Senior Design Project - II (FYDP-II)	0	3	**16**
	MINE-317 / MINE-318 & L	Underground Mine Design / Mine Hazards & Safety	3 / 3	0 / 1	
	MGMT-211 / MINE-408 & L	Engineering Management / Tunnel Engineering	2 / 3	0 / 1	
	—	First Aid Course	0	0	

2. Master of Science Program (MS)

Admission Requirements

- **Academic Background:** 16 years of education or an equivalent benchmark qualification (such as a 4-year BS or B.E. degree) in a relevant engineering discipline from an HEC-recognized university with at least **60% marks** (under the annual grading setup) or a minimum CGPA of **2.5 out of 4.0**. *(Note: The raw web copy contains a localized placeholder text errantly listing Chemical/Petroleum fields)*.
- **Testing:** A valid **GAT General** score with a passing benchmark of at least **50% marks**.

Degree Requirements

- **Total Credit Hours:** 30
- **Total Coursework Requirements:** 8 advanced postgraduate courses.
- **Minimum Academic Standing:** CGPA $\ge$ 2.5